Exercise 8

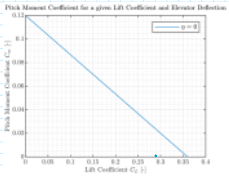
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Exercise 8: Trim

A stable aircraft configuration shows the following characteristics:

Lift coefficient at zero AoA	C_{L0}	0.3
Pitch moment coefficient at zero AoA	C_{m0}	0.02
Elevator pitch moment derivative	$C_{m\eta}$	+-2.5
Elevator lift derivative	$C_{L\eta}$	+-0.5
Aircraft mass	m	2000 kg
Altitude	h	3000 m
Airspeed in Mach	M	0.2
Reference area	S_{ref}	20 m²

Note: at this point we don't know the sign of the derivatives. That's why their sign is denoted by „+-". The following relation between the lift- and the moment-coefficient (for one particular elevator deflection!) is known:



a. What is the correct sign of $C_{m\eta}$?

$M_{\eta+}$ ≙ Moment due to positive η deflection
$M_{\eta-}$ ≙ Moment due to negative η deflection
$\Rightarrow C_{m\eta} < 0$



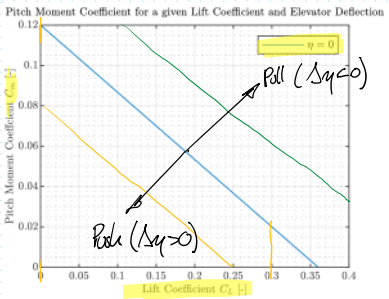
positive deflection
↳ negative Reaction moment

$C_{x|y}$
Change of C_x due to y

$C_{L\alpha}$ ≙ Change of coefficient C_L of Lift due to Angle of Attack α
C_{l_p} ≙ Change of coefficient C_l of Rollmoment due to roll rate p ≙ damping
C_{m_q} ≙ Change of pitchmoment coefficient due to pitch rate
C_{l_β} ≙ Change of coefficient of rollmoment due to a change in sideslip β

b. How does the relation/picture (qualitatively) change when the pilot pulls/pushes the stick?

↳ $\eta > 0, \eta < 0$



$C_L = C_{L0} + C_{L\alpha} \cdot \alpha + C_{L\eta} \cdot \eta \rightarrow \alpha = \frac{C_L - C_{L0} - C_{L\eta}\eta}{C_{L\alpha}}$ (I)

$C_m = C_{m,0} + C_{m\alpha} \cdot \alpha + C_{m\eta} \cdot \eta$ (II)

(I) in (II):

$C_m = C_{m0} + C_{m\alpha} \cdot \frac{C_L - C_{L0} - C_{L\eta}\eta}{C_{L\alpha}} + C_{m\eta} \cdot \eta$

$C_m = C_{m0} + (C_L - C_{L0} - C_{L\eta}\eta) \cdot \frac{C_{m\alpha}}{C_{L\alpha}} + C_{m\eta}\eta$

$C_m = C_{m0} + C_L \cdot \frac{C_{m\alpha}}{C_{L\alpha}} - \frac{C_{L0} C_{m\alpha}}{C_{L\alpha}} - \frac{C_{L\eta}\eta \cdot C_{m\alpha}}{C_{L\alpha}} + C_{m\eta}\eta$

$C_m = C_{m0} + C_L \cdot \frac{C_{m\alpha}}{C_{L\alpha}} - C_{L0}\frac{C_{m\alpha}}{C_{L\alpha}} + \eta\left(C_{m\eta} - C_{L\eta}\frac{C_{m\alpha}}{C_{L\alpha}}\right)$

$\frac{C_{m\alpha}}{C_{L\alpha}}$ = Slope

Results:

1. Slope is independent of η as $C_{m\alpha}$ and $C_{L\alpha}$ are const.
   ⇒ Parallel displacement depending on η

2. In what direction will the curve be displaced depending on η?

look at Plot

Subst. a) $C_{m\eta} - C_{L\eta}\frac{C_{m\alpha}}{C_{L\alpha}}$? ⇒ $\frac{C_{m\alpha}}{C_{L\alpha}} = \frac{\frac{\partial C_m}{\partial \alpha}}{\frac{\partial C_L}{\partial \alpha}} = \frac{\partial C_m}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial C_L} = \frac{\partial C_m}{\partial C_L} = -\frac{0.12}{0.36} \approx -0.33$

$C_{m\eta} < 0$

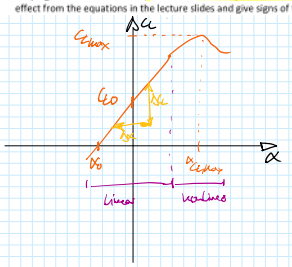
↳ have a look at aerodynamics: $C_{L\eta} > 0$

≙ SM must be stable aircraft if it is negative

3. Put all together: $C_{m\eta} - C_{L\eta}\frac{C_{m\alpha}}{C_{L\alpha}} = -2.5 - (0.5 \cdot (-0.33)) = -2.335 < 0$

⇒ positive η leads to negative displacement of the curve $\frac{C_m}{C_L}$

c. What difference does the curve „**Lift coefficient over angle of attack**" show, if instead of keeping the elevator constant, the elevator is always moved to the trim condition? Derive this effect from the equations in the lecture slides and give signs of the involved derivatives.



Trim Condition: $C_m(\eta_{trim}) = 0$

$C_L = C_{L0} + C_{L\alpha} \cdot \alpha + C_{L\eta} \cdot \eta$

$C_m = C_{m,0} + C_{m\alpha} \cdot \alpha + C_{m\eta} \cdot \eta$

↓ include as trim condition

$C_{L_{trim}} = C_{L0} + C_{L\alpha} \cdot \alpha_{trim} + C_{L\eta} \cdot \eta_{trim}$ (I)

$0 = C_{m,0} + C_{m\alpha} \cdot \alpha_{trim} + C_{m\eta} \cdot \eta_{trim} \rightarrow \eta_{trim} = \frac{-C_{m0} - C_{m\alpha} \cdot \alpha_{trim}}{C_{m\eta}}$ (II)

$$C_{L,trim} = C_{L0} + C_{L\alpha} \cdot \alpha_{trim} + C_{L\eta} \cdot \eta_{trim} \quad (I)$$

$$0 = C_{m,0} + C_{m\alpha} \cdot \alpha_{trim} + C_{m\eta} \cdot \eta_{trim} \rightarrow \eta_{trim} = \frac{-C_{m0} - C_{m\alpha} \cdot \alpha_{trim}}{C_{m\eta}} \quad (II)$$

Insert (II) in (I):

$$C_{L,trim} = C_{L0} + C_{L\alpha} \cdot \alpha_{trim} + C_{L\eta} \cdot \left(\frac{-C_{m0} - C_{m\alpha} \cdot \alpha_{trim}}{C_{m\eta}} \right)$$

$$C_{L,trim} = \left(C_{L0} - \frac{C_{m0} \cdot C_{L\eta}}{C_{m\eta}} \right) + \alpha_{trim} \left(C_{L\alpha} - \frac{C_{L\eta} \cdot C_{m\alpha}}{C_{m\eta}} \right)$$

Compare to $C_L = C_{L0} + C_{L\alpha} \cdot \alpha$

correct C_{L0} for the trim — correct $C_{L\alpha}$ for trim

$$C_{L,trim} = C_{L0}^* + \alpha_{trim} \cdot C_{L\alpha}^*$$

$C_{m0} > 0$ given

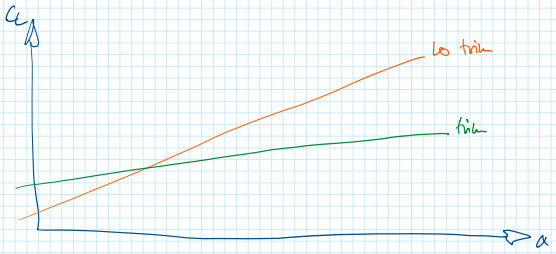
$C_{L\eta} > 0$ } Schaale a)

$C_{m\eta} < 0$

$C_{m\alpha} < 0 \rightarrow$ stable aircraft

$\Rightarrow C_{L0,trim} = C_{L0}^* > C_{L0}$ — larger intercept

$\Rightarrow C_{L\alpha,trim} = C_{L\alpha}^* < C_{L\alpha}$ — smaller slope



d. Calculate the required elevator deflection for a steady-state horizontal cruise flight. Hint: You can assume the standard atmosphere.

$$\eta_{trim} = \frac{C_{L,trim} - C_{L0} + \frac{C_{m0} \cdot C_{L\alpha}}{C_{m\alpha}}}{\frac{C_{L\alpha} \cdot C_{m\eta}}{C_{m\alpha}} - C_{L\eta}}$$

unknown — known — known — known — known — known — known

Derivation:

$$\alpha_{trim} = -\frac{C_{m0}}{C_{m\alpha}} - \frac{C_{m\eta}}{C_{m\alpha}} \cdot \eta_{trim}$$

$$C_{L,trim} = C_{L0} + C_{L\alpha} \cdot \alpha_{trim} + C_{L\eta} \cdot \eta_{trim}$$

Obtaining $C_{L,trim}$: $L = mg$

$$\frac{1}{2} \rho v^2 S_{ref} \cdot C_{L,trim} = mg \rightarrow C_{L,trim} = \frac{mg}{\frac{1}{2} \rho V_{TAS}^2 S_{ref}}$$

unknown!

Calculate density:

$$\rho = \rho_{ISA} \left(1 + \frac{\gamma_{TR}}{T_{ISA}} \cdot H_G \right)^{\frac{1}{\mu - 1}} \quad \text{with } H_G = \frac{r_E \cdot h}{r_E + h} = 2998.6\,m$$

$$\rho = 0.903 \frac{kg}{m^3}$$

Calculate V_{TAS}:

$$Ma = 0.2 \rightarrow V_{TAS} = Ma \cdot a = Ma \cdot \sqrt{\kappa R T} = 0.2 \cdot \sqrt{1.4 \cdot 287.15 \cdot 268.66}\ m/s$$

$$T = T_{ISA} \left(1 + \frac{\gamma_{TR}}{T_{ISA}} \cdot H_G \right) = 268.66\,K \qquad V_{TAS} = 65.73\ m/s$$

$\Rightarrow$ Insert and solve for $C_{L,trim}$: $C_{L,trim} = 0.5$

$$\eta_{trim} = -0.0027\,rad \approx -0.1547°$$